

INSERTION SORT:

Insertion Sort is a simple comparison-based sorting algorithm. It works by taking one element at a time and inserting it into its correct position in the already sorted part of the array. It is similar to arranging playing cards in your hand.

Algorithm:

1. Read the number of elements.
2. Enter the array elements.
3. Assume the first element is already sorted.
4. Pick the next element as the **key**.
5. Compare the key with the elements before it.
6. Shift all larger elements one position to the right.
7. Insert the key into its correct position.
8. Repeat the process until all elements are sorted.
9. Display the sorted array.

Time Complexity:

Case	Time Complexity
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Best Case	$O(n)$
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Average Case	$O(n^2)$
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Worst Case	$O(n^2)$
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Space Complexity:

Type	Space Complexity
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Insertion Sort	$O(1)$
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Output:

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Enter the number of elements:
6
Enter the elements:
67
45
90
32
48
2
Sorted array:
2
32
45
48
67
90

Process finished with exit code 0
```